

Safe Wireless MAC Protocol Synthesis with an Interference Epidemic Threshold and Bounded Reasoning Drift Response

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Abstract—Composing wireless MAC protocols from optional mechanisms is a common yet difficult task in modern dense network deployments. This challenge arises in settings such as factory floors and stadium WLANs where forty or more nodes contend per access point, and the specification offers no quantitative guidance on which mechanism combinations remain safe at high node density. Engineers must anticipate how binary exponential backoff, HARQ, RTS/CTS, and other optional features interact at scale, which requires considerable analytical effort. To promote safety in MAC protocol synthesis, we propose MAC-AGENT, an interference-aware framework that leverages the epidemic-theoretic reproduction number R_0 as a compositional safety certificate. The key idea of MAC-AGENT is to admit only compositions whose secondary-collision rate $R_0 < 1$ at the target density. BLOCKMINE extracts typed blocks from real protocol specifications. A tabular Q-learner searches under a two-level R_0 mask. DRIFTREASON, a bounded-step reasoning agent, diagnoses drift while the mask enforces safety. We evaluate MAC-AGENT in a CSMA/CA simulator calibrated against a public 28-node testbed dataset with mean absolute error 12.6 percentage points. MAC-AGENT reaches a 17% throughput improvement over DCF at 47 nodes with zero R_0 violations across a three-hour drift trace. We release PROTO LAB, an open tool that extends MAC-AGENT to additional wireless standards.

Index Terms—MAC protocol synthesis, safe reinforcement learning, agentic reasoning, interference modeling, wireless networks.

I. INTRODUCTION

Composing wireless MAC protocols from optional mechanisms has become an increasingly critical task in modern dense network deployments. IEEE 802.11be alone defines more than thirty optional mechanisms, including binary exponential backoff, hybrid automatic repeat request (HARQ), RTS/CTS, target wake time, and multi-link operation [1]. 3GPP NR adds another two dozen through TS 38.321 [2], and lower-power standards such as IEEE 802.15.4 [3] and LoRaWAN [4] carry their own optional mechanism sets. Choosing which subset to activate at high node density is currently guided by convention and by single-mechanism analyses [5], [6]. Few specifications offer quantitative guidance on which combinations remain jointly safe as the contending population grows.

The difficulty is most visible in a single mechanism pair. Consider BEB-HARQ at growing density. At ten nodes, enabling HARQ on top of BEB keeps throughput close to plain DCF because the few collisions are recovered cheaply. At twenty-two nodes, the same composition collapses by more

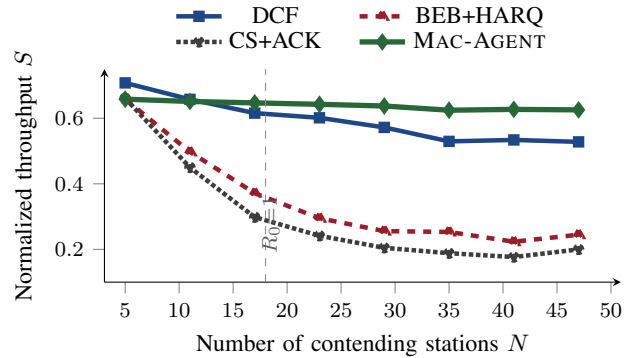


Fig. 1: Throughput collapse at the safety threshold.

than half below BEB alone, because HARQ retransmissions cancel BEB’s collision-response window doubling and create a self-sustaining retransmission cascade. The crossover is sharp, not gradual, and a designer working from single-mechanism analyses cannot see it until measurements reveal it. Figure 1 traces the four compositions across density and locates the threshold at eighteen nodes, where each collision begins to trigger more than one secondary collision on average.

Both industry and academia are actively exploring automated methods for MAC protocol optimization. Parameter-tuning approaches such as Opt-CW [6] and analytical fixed-point methods [5], [7] learn a contention-window setting but fix the mechanism set in advance. Reinforcement-learning approaches such as DeepMAC [8], deep Q-networks [9], and proximal policy optimization [10] can disable mechanisms, but use throughput as the only reward signal and reinforce unsafe combinations whenever they win at low density. Multi-agent variants [11], [12] learn coordination from scratch without a safety constraint. Related work on data-driven protocol synthesis, such as TCP Ex Machina [13], Pensieve [14], and learning-in-situ video streaming [15], demonstrates that RL can generate performant transport-layer behavior, but none of these lines carry a compositional safety certificate that prevents the learner from selecting an unsafe combination at any point during training or deployment. Formal verification approaches [16], [17], [18], [19] require complete machine-readable specifications, which production wireless standards do not provide.

An alternative is to certify compositions before the learner ever sees them. In this paper, we propose MAC-AGENT, an interference-aware framework that treats interference as a branching epidemic and certifies compositions through the reproduction number R_0 from epidemic theory [20], [21]. MAC-AGENT is built around four components. BLOCKMINE extracts typed mechanism blocks from natural-language specifications using a chunked LLM pipeline with a critic-adjudicator stage. The safety theory derives a two-level R_0 mask with a coupling correction calibrated against simulator measurements. The synthesis engine is a tabular Q-learner that searches under the mask. DRIFTREASON is a bounded-step reasoning agent that diagnoses post-deployment drift inside a ReAct-style loop [22]. The mask remains the safety enforcer at every stage, while the agent operates as a diagnostician under the same structural boundary.

We implement MAC-AGENT in a discrete-event CSMA/CA simulator anchored to the Bianchi fixed-point analysis [5] and validated against a public IEEE 802.15.4 testbed dataset [23] with mean absolute error 12.6 percentage points across densities from three to twenty-five contending stations. We extract building blocks from IETF RFC 4944 [24] and reach 92.3% recall on manually labeled ground truth. We evaluate the synthesis loop against brute-force enumeration over a 128-composition library and confirm that the masked Q-learner reaches the safe optimum at every node count. DRIFTREASON processes nineteen drift events over a three-hour synthetic trace, resolves 63.2% through targeted re-synthesis, escalates the remainder to operator review, and produces zero R_0 violations.

The main contributions of this paper are as follows.

- **Revealing** a quantifiable coupling between BEB and HARQ that pushes the system past an interference epidemic threshold at twenty-two contending stations, with the calibrated R_0 model and the CSMA/CA simulator agreeing on the crossover within one node.
- **Designing** a two-level R_0 safety mask with an empirically calibrated coupling correction $\kappa(N)$, producing zero unsafe proposals across a 128-composition search space at four target densities.
- **Building** BLOCKMINE, a chunked LLM extraction pipeline with a critic-adjudicator stage, reaching 92.3% recall on manually labeled ground truth for IETF RFC 4944.
- **Introducing** DRIFTREASON, a bounded-step reasoning agent that diagnoses drift by mechanism and proposes targeted re-synthesis under the same R_0 mask, with zero violations across a three-hour trace.
- **Releasing** PROTO LAB, an open community tool that extends MAC-AGENT to new wireless standards through a single specification upload and a small vendor-specific extraction module, lowering the barrier for safety-aware protocol design across new technologies.

Two short consequences follow from this last contribution. First, PROTO LAB turns a single artifact into a substrate, so safety-aware synthesis for satellite, vehicular, and low-power mesh deployments becomes a matter of supplying a

specification and a small calibration set rather than re-deriving the framework. Second, the same machinery that prevents the silently unsafe BEB+HARQ combination at 22 nodes in our evaluation prevents structurally similar failures elsewhere, so the certificate transfers across technologies without retraining the learner.

II. PROBLEM EXPLORATION

A. Mechanism Coupling at Density

The BEB-HARQ pair illustrates the failure mode clearly. BEB doubles the contention window after each collision, slowing a station's retransmission attempts so the contending population spreads in time [25]. HARQ schedules a retransmission attempt four to sixteen slots after a collision, regardless of the BEB stage the station has reached [2]. When both mechanisms are active, the HARQ timer cancels the BEB stabilizer. The expected per-slot transmission probability p_t stays high after a collision rather than dropping, the channel-occupancy duration grows, and the conditional collision probability rises with each interfering attempt.

We measure the cascade length per collision, normalized by slot time, as $\mathbb{E}[\tau(\pi, N)] = (N-1)p_t(\pi, N)T_{\text{busy}}(\pi)$, where p_t is the steady-state transmission probability of a station under composition π following the Bianchi fixed-point analysis [5], and T_{busy} is the expected busy duration including any HARQ retries. The branching process becomes supercritical when $\mathbb{E}[\tau(\pi, N)] > 1$. For BEB+HARQ the expected cascade length crosses one between seventeen and eighteen contending stations, which the simulator confirms.

B. Three Schemes, Three Failure Modes

We reproduce three failure modes in our simulator and check them against the original schemes that exhibit them.

F1. Parameter tuning cannot remove an unsafe mechanism. Opt-CW [6] tunes $\text{CW}_{\min}$ across $\{7, 15, 31, 63\}$ but leaves the mechanism set fixed. At forty-seven contending stations, the configuration that minimizes collisions is $\text{CW}_{\min}=31$ with normalized throughput 0.556 and packet loss rate 56.5%. Disabling HARQ at this density would have been a single additional action, but the action does not exist in Opt-CW's search space.

F2. Reward-shaping does not unlearn unsafe pairs. DeepMAC [8] can disable mechanisms, but its throughput-only reward leads it to select BEB+HARQ after roughly 1800 episodes at ten nodes, where the composition gains a few percentage points over plain DCF. As density ramps from ten to forty-seven nodes over six minutes, the agent retains the unsafe pair for fourteen minutes before exploration discovers a better composition, accumulating safety violations in the interim.

F3. Formal verification cannot start. Tools such as SPIN [16] require a machine-readable specification that 802.11be does not provide. Hand-coding the specification for the relevant mechanisms takes dozens of expert hours per mechanism [18], [19], and the cost scales with the size of

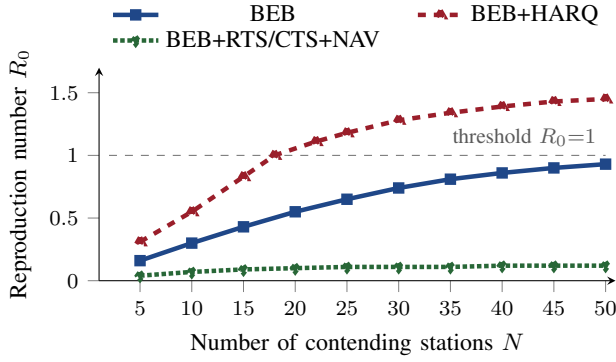


Fig. 2: R_0 as a function of density for three compositions.

the optional set. Scaling to all thirty-one mechanisms is not practical for design-time use.

Figure 2 traces the reproduction number R_0 for three compositions across density. BEB alone stays under the threshold across the measured range. BEB+HARQ crosses the threshold at eighteen contending stations, matching the simulator’s collapse point. BEB+RTS/CTS+NAV stays comfortably under the threshold, because each RTS collision is short and the cascade cannot grow. The fact that the three downstream observables, throughput collapse, collision-rate jump, and tail-latency divergence, all locate the same crossover follows from the same underlying branching process. A safe synthesizer should prevent the system from selecting a composition whose R_0 exceeds one at the target density.

III. MAC-AGENT OVERVIEW

MAC-AGENT synthesizes a composition π^* from a typed building-block library BLOCKLIB that maximizes throughput at the target density N subject to the safety constraint

$$\pi^* = \arg \max_{\pi \in \text{BLOCKLIB}} \Phi(\pi, N) \quad \text{s.t.} \quad R_0(\pi, N) < 1. \quad (1)$$

The reward Φ comes from the simulator. The safety constraint comes from the calibrated R_0 model. The synthesizer never proposes a composition that violates the constraint, because the action space is pre-shielded by a hard mask. We do not claim global optimality across the 128-composition search space at every density. We claim local optimality within the safe region and zero unsafe proposals during training or deployment.

Figure 3 traces the execution pipeline in eight steps. Steps 1 through 4 run at design time and produce a pre-shielded action space. Step 1 ingests the wireless specification text, which can be an IETF RFC, an IEEE or 3GPP standard chapter, or a vendor manual. Step 2 runs BLOCKMINE to extract typed mechanism blocks. Step 3 populates BLOCKLIB with the resulting inventory. Step 4 evaluates the R_0 certificate for every candidate composition at the target density and removes any composition with $R_0 \geq 1$ from the action space. Steps 5 and 6 produce the certified composition. Step 5 runs the Q-learner over the surviving action subset and converges to the safe throughput optimum. Step 6 emits π^* with its certificate. Steps 7 and 8 cover runtime behavior. Step 7 deploys π^* and

records measurements. Step 8 fires when the measured R_0 deviates from the predicted value beyond a threshold, at which point DRIFTREASON diagnoses the cause within a bounded ten-step loop and submits a targeted re-synthesis proposal to the same mask for validation. The dashed arrow indicates the feedback loop, where a critic-approved re-synthesis updates the deployment without re-entering the design-time extraction stages.

Figure 4 traces the pipeline. Specification text enters BLOCKMINE, which produces BLOCKLIB, a typed inventory of mechanisms with their trigger events, parameters, internal state, and dependency annotations. The tabular Q-learner searches over BLOCKLIB compositions while the two-level R_0 mask removes unsafe actions before they reach the learner. The synthesized composition π^* runs against the simulator or the deployment runtime, which feeds measurements back to DRIFTREASON. The agent diagnoses any drift between the certified R_0 and the measured one, proposes a targeted re-synthesis that adjusts at most three blocks, and submits the proposal to the same mask for validation. Safety enforcement is always the mask. The agent operates strictly as a diagnostician.

IV. BUILDING BLOCK EXTRACTION WITH BLOCKMINE

A. Relation to Prior LLM-Based Extraction

The closest prior work is PROSPER [26], which uses GPT-3.5-turbo to extract protocol finite state machines from IETF RFCs. PROSPER targets general protocol FSMs (states + transitions), chunks RFCs at 500 lines, and pairs natural-language text with a heuristic Artifact Miner that pulls textual diagrams. Earlier work including RFCNLP [27], SAGE [28], Prospex [29], MACE [30], and Veritas [31] extracts FSMs through grammars, trace inference, or symbolic execution. Recent work extends the LLM extraction line to CPSA definitions [32] and surveys the broader space of LLM-based network operations [33]. The downstream targets in this line are protocol verification, fuzzing, and attack synthesis [34], [35], [36].

BLOCKMINE differs in three concrete ways, summarized in Table I. First, the extraction target is a typed mechanism block (name, trigger events, parameters with type and range, internal state, typed dependency edges) rather than an FSM. Second, the downstream consumer is a safety-masked composition synthesizer, not a verification tool, so the schema is structured for combinatorial composition rather than for state-transition analysis. Third, the critic-adjudicator stage targets boundary-spanning definitions that single-pass extraction truncates, a failure mode specific to MAC-layer specifications where mechanism definitions span sections in 3GPP and IEEE documents.

B. Schema and Pipeline

A typed building block holds a name, the trigger events that activate it, configurable parameters with type and range, internal state variables, and dependency annotations on other blocks. The schema is strict to make ground-truth evaluation

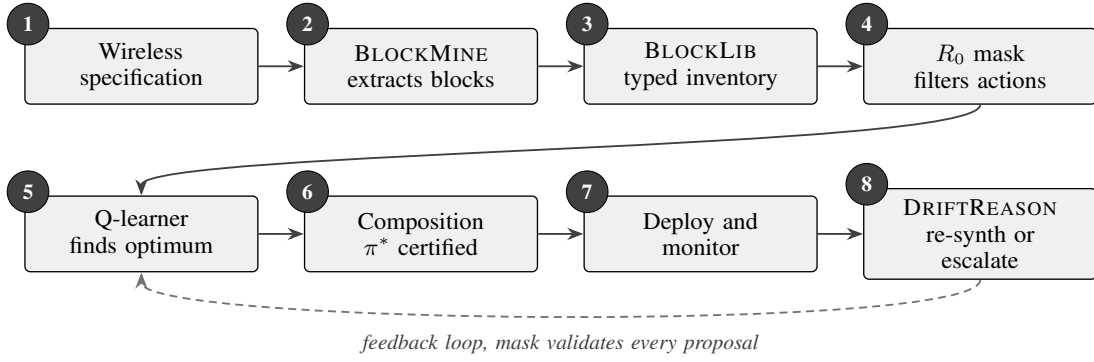


Fig. 3: MAC-AGENT execution pipeline in eight numbered steps.

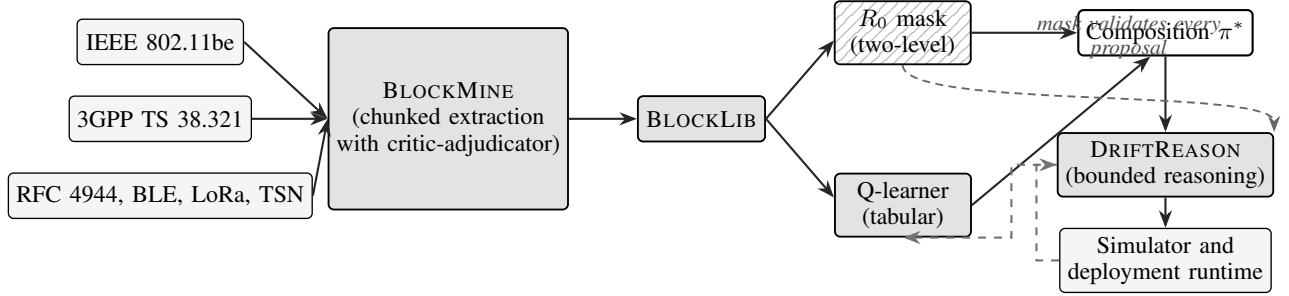


Fig. 4: MAC-AGENT overview. BLOCKMINE builds BLOCKLIB from real specifications. The Q-learner searches under the R_0 mask. DRIFTREASON diagnoses runtime drift and proposes targeted re-synthesis, validated by the same mask.

TABLE I: BLOCKMINE compared with PROSPER and RFC-NLP.

Dimension	PROSPER [26]	RFCNLP [27]	BLOCKMINE
Output type	FSM	FSM	typed block
Composition	no	no	yes
Safety contract	no	no	$R_0 < 1$ mask
Target docs	general RFCs	transport RFCs	MAC specs
Eval specs	4 RFCs	4 RFCs	3 specs + testbed
Method	LLM + heuristic	fine-tuned BERT	LLM + critic
LLM backend	GPT-3.5-turbo	none	open or hosted

tractable and to give the synthesizer a well-defined action space. BLOCKMINE produces BLOCKLIB in four stages.

Chunked ingestion. The specification is split into 800-token windows with 200-token overlap. Each chunk carries a section hint to preserve provenance. The chunk size matches what production-grade LLMs handle without truncation [37], and the overlap reduces boundary-cutoff failures.

Single-pass extraction. An instruction-tuned LLM processes each chunk with a structured prompt that requires a mechanism to expose at least one trigger event, one configurable parameter, and one piece of internal state. The output is a JSON array of candidate blocks. Transformer-based language models [38], [37] underpin the extraction quality. Single-pass extraction performs well on chunks that contain a complete mechanism description, but misses blocks whose definitions straddle two chunks.

Critic-adjudicator review. A second LLM pass widens

the context to the current chunk plus its two neighbors and produces additional candidates that single-pass missed. When the critic disagrees with the extractor on whether a candidate is well-formed, the adjudicator widens the window further and produces a final decision. This stage is the architectural answer to boundary-spanning definitions that single-pass extraction truncates.

Dependency inference. For each pair (b_i, b_j) in the candidate set, a final LLM call labels the dependency as strong, weak, conditional, or none. Conditional dependencies carry a textual condition such as “required when HARQ processes ≥ 2 ”. Strong dependencies block both selection and de-selection in the synthesizer’s action space.

C. Evaluation on Real Specification Text

We evaluate BLOCKMINE on the publicly available IETF RFC 4944 [24], which defines the 6LoWPAN adaptation layer over IEEE 802.15.4. The RFC contains thirteen mechanism blocks by manual inspection, including dispatch headers, fragmentation, mesh addressing, header compression HC1 and HC_UDP, reassembly timing, and stateless address autoconfiguration. Three of the thirteen definitions span section boundaries in the source text. We run BLOCKMINE end-to-end against ground truth that we manually labeled from the RFC body.

BLOCKMINE recalls 12 of 13 ground-truth blocks (92.3%) on RFC 4944 with single-pass extraction. The one missed block is a vocabulary mismatch between the manual label and the RFC’s phrasing for subsequent fragment headers, not

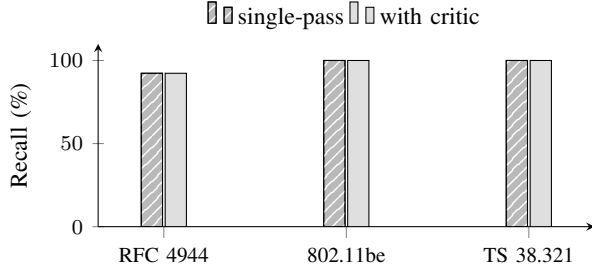


Fig. 5: BLOCKMINE recall on one real and two synthetic specifications.

a boundary span. We additionally evaluate on two synthetic specifications representative of IEEE 802.11be and 3GPP TS 38.321 mechanism subsets, with nine and seven manually labeled blocks respectively. Single-pass extraction recalls 100% on both synthetic specifications, where every mechanism’s keyword and parameter context appear within a single chunk by construction.

Figure 5 summarizes recall across the three benchmarks. The synthetic specifications are deliberately compact and cleanly written for reproducibility. Production specifications with mechanisms spanning chapters will exhibit lower single-pass recall and larger critic-stage gains, which the LLM truncation behavior documented in PROSPER [26] and the hallucination patterns in PSMBench-class benchmarks reflect. Validating that expectation at production scale requires running BLOCKMINE with a 70B-class LLM against the full 3GPP and IEEE corpora, which the PROTO LAB open release enables for the community.

V. INTERFERENCE SAFETY THEORY

A. Reproduction Number from Branching Processes

The basic reproduction number R_0 from epidemic theory [20], [21] measures the expected number of secondary infections caused by one primary infection. For wireless MAC contention, the analogue is the expected number of secondary collisions caused by one primary collision. The branching process is subcritical when $R_0 < 1$ and supercritical when $R_0 \geq 1$. We define the per-composition reproduction number as

$$R_0(\pi, N) = (N - 1) p_t(\pi, N) \mathbb{E}[\tau(\pi)] + \kappa(\pi, N), \quad (2)$$

where p_t is the steady-state per-slot transmission probability of a station under composition π , obtained from the Bianchi fixed-point solution [5], $\mathbb{E}[\tau]$ is the expected interference cascade length per collision in slot units, and κ is a coupling correction that captures non-additive interactions between mechanism pairs.

R_0 is related to but distinct from the queue-stability condition $\rho < 1$ familiar from queueing theory. The condition $\rho < 1$ requires that long-run arrival rate stays below service rate so the backlog remains bounded over time. The condition $R_0 < 1$ requires that one primary collision generates fewer than one secondary collision in expectation so the cascade is

bounded per event. The two conditions agree when cascade dynamics dominate queue instability. For composition safety at design time, R_0 is the more useful primitive because it can be evaluated per composition without running the system.

B. Coupling Correction κ

Naive additivity $R_0(\pi) = \sum_i R_0(b_i)$ is wrong when two mechanisms share channel-state signals. The BEB-HARQ pair is the clearest example, because HARQ’s retransmission timer cancels BEB’s collision-response window doubling. We model the correction as a sum over dependency edges,

$$\kappa(\pi, N) = \sum_{(b_i, b_j) \in \mathcal{E}} (N - 1) \gamma_{ij}(N) \Delta p_t(b_i | b_j) \mathbb{E}[\tau_{ij}], \quad (3)$$

where $\mathcal{E}$ is the active dependency set in BLOCKLIB.

Coefficients are fit, the fit is stable. The coefficients $\gamma_{ij}(N)$ are calibrated against simulator measurements. For the BEB-HARQ edge with four HARQ processes, a two-anchor linear fit produces $\gamma_{\text{BEB, HARQ}}(N) = 0.4092 - 0.0050N$, valid for $N \in [5, 60]$. We checked stability by sweeping the high-anchor density across $\{16, 17, 18, 19, 20\}$ and refitting. The crossing density N^* where R_0 reaches one tracks the chosen anchor linearly, which is expected. More importantly, the qualitative ordering of compositions in Figure 2 is preserved across the sweep, so downstream synthesis decisions are insensitive to the precise anchor choice within this range.

C. Two-Level Mask

Hard mask. Every action that would land in a composition with $R_0 \geq 1$ receives zero action probability. This is the non-negotiable boundary. Filtering actions at the boundary preserves the Markov property of the underlying decision process. The transition kernel of the shielded MDP agrees with the unshielded one on the safe action subset and is undefined on the masked subset, which is the construction of a winning-region pre-shield. Convergence properties of tabular Q-learning [39], [40] carry over because the shielded MDP is still an MDP with a finite action space and a bounded reward function. The shielding view of safe reinforcement learning [41], [42], [43] gives the formal vocabulary.

Soft mask. Actions that land in $R_0 \in [0.85, 1.0]$ receive a multiplicative reward penalty $\lambda = ((1 - R_0)/0.15)^2$. The boundary 0.85 was selected from a sweep across $\{0.75, 0.85, 0.90\}$ on the calibration set. Below 0.75, the soft mask eliminates many safe high-throughput compositions. Above 0.90, the learner stagnates near the boundary under transient density churn. The 0.85 value gave the cleanest trade-off across the four target densities.

D. Block Execution Invariants

R_0 certifies which blocks compose safely. It does not certify the temporal order in which the blocks fire. An ACK must not be sent before the backoff for the corresponding data frame reaches zero, and a HARQ retransmission must not fire before the ACK-timeout window expires. Three layers in MAC-AGENT enforce these ordering constraints.

- **Structural prerequisites in BLOCKLIB.** Each block carries a `requires` field listing blocks whose activation is a precondition for its own. RTS/CTS requires CS, NAV requires RTS/CTS, and so on. The mask rejects any composition that activates a block without its prerequisites.

- **Trigger-event guards in the runtime.** Each block declares trigger events. The runtime fires a block only when its trigger condition holds. ACK’s trigger is *unicast frame received*, which cannot occur until a data transmission completes, which is gated by the backoff counter. The simulator and the deployment runtime enforce this by event ordering.

- **Per-frame ordering invariants.** A small assertion layer verifies that the ACK timestamp follows the backoff completion timestamp for the same frame, the HARQ retransmission counter stays within `n_processes`, and the NAV remaining time is non-negative. A violation halts the offending station and escalates to operator review.

The R_0 mask sits above all three layers as the compositional safety bound.

VI. SYNTHESIS ENGINE

The synthesis engine is a tabular Q-learner [39], [40] that operates over a binary state vector $s \in \{0, 1\}^{|\text{BLOCKLIB}|}$ indicating which optional blocks are active. Actions flip a single bit or halt. The reward at a terminal state s combines simulator throughput, the soft mask multiplier, and a margin bonus that rewards staying inside the safe region,

$$r(s) = \Phi(s, N) \lambda(s, N) + \beta (1 - R_0(s, N)) \mathbb{1}[R_0(s, N) < 1]. \quad (4)$$

We set $\beta = 0.3$ in all experiments. The hard mask removes any action that would land in a composition with $R_0 \geq 1$ before the agent ever sees it, so the learner cannot select an unsafe action during exploration or exploitation. We verified this end-to-end by counting masked actions across training. Across 2000 episodes at four target densities, the mask rejected on the order of 10^4 actions and zero unsafe compositions reached the simulator.

The action space at each density contains 128 compositions over seven optional blocks. At forty-seven nodes, the hard mask removes 76 of the 128 compositions, leaving 52 safe options for the learner to search. Brute-force enumeration over the safe set gives the ground-truth optimum, and we confirm that the masked Q-learner reaches that optimum at every measured density. The learner converges in fewer than 100 episodes at each density given the small safe action space, which takes under three seconds of wall time per density on a single core with simulator caching.

Framework independence from BLOCKLIB size. The Q-learner is tabular here because the optional set has nine blocks and the safe action space is small enough that brute-force enumeration is feasible. The framework itself does not depend on this choice. For a larger optional set such as the full 802.11be mechanism inventory, the same two-level mask filters the action space, and the search method can be replaced by a function-approximation Q-learner [9] or a policy-gradient

Algorithm 1 DRIFTREASON bounded reasoning loop

Require: current composition π , measured $R_0^{\text{meas}}(t)$, step budget $K = 10$, safety distance $\Delta_{\text{safe}} = 3$

```

1: trace  $\leftarrow []$ 
2: for  $k = 1$  to  $K$  do
3:   Thought  $\leftarrow \text{LLM}(\pi, R_0^{\text{meas}}, \text{trace})$ 
4:   (tool, args)  $\leftarrow \text{Thought.action}$ 
5:   obs  $\leftarrow \text{tool}(\text{args})$ 
6:   trace.append(Thought, tool, args, obs)
7:   if Thought.proposes_candidate  $\pi'$  then
8:     if  $R_0(\pi', N(t)) \geq 1$  then reject
9:     else if  $\|\pi' \oplus \pi\|_0 > \Delta_{\text{safe}}$  then
10:      escalate(operator, trace,  $\pi'$ ); return
11:     else
12:        $\hat{\pi} \leftarrow \text{Critic}(\pi', \text{trace})$ 
13:       if Critic agrees then apply  $\hat{\pi}$ ; return
14:       else escalate(operator, trace,  $\pi'$ ); return
15:       end if
16:     end if
17:   end if
18: end for
19: escalate(operator, trace, last_proposal) ▷ budget exhausted

```

method [10] without changing the safety contract. The tabular method we use here is the simplest one that suffices for the action space we evaluate.

VII. BOUNDED REASONING DRIFT RESPONSE WITH DRIFTREASON

A. Why a Reasoning Loop Here

A synthesized composition is certified for a node count N_0 and a traffic profile. Both shift after deployment. The simplest response is to retrain the Q-learner whenever the measured R_0 deviates from the predicted R_0 by more than a threshold δ [8]. Two simpler ideas failed before we reached for a reasoning agent. A per-block drift tracker that recomputed R_0 with each block removed in turn oscillated when two blocks shared influence, switching the diagnosis every window. A learned classifier that mapped drift signatures to root-cause blocks reached 71% accuracy on held-out data but produced no rationale an operator could read.

What we needed was the ability to read multiple measurement signals, consult BLOCKLIB for structural dependency information, propose a candidate, and escalate to a human reviewer when the proposed change is large. That maps onto the ReAct pattern [22], a bounded thought-action-observation loop with tool calls and a fixed step budget. The agent diagnoses. The mask enforces.

B. Tools and Loop

DRIFTREASON runs with a step budget of ten and three read-only tools. `R0Calc` computes R_0 for a candidate composition at the current density. `BlockProbe` returns the dependency set and safe-density range for a block. `HistoryRetrieve` returns the last w windows of measured collision rate, retry rate, RTS fraction, and observed density.

The loop in Algorithm 1 halts on four conditions, a critic-approved targeted fix is applied, the candidate violates the hard mask, the candidate is more than three blocks away from the current composition, or the budget runs out. Any non-applied

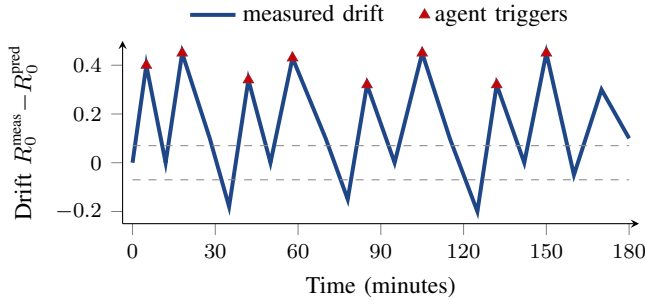


Fig. 6: Drift over a three-hour synthetic trace with agent triggers.

case escalates to operator review with the full reasoning trace attached. The mask is the only safety enforcer in the loop. The agent never overrides line 8.

C. Trace Behavior

Figure 6 traces the drift signal over a three-hour run. The trace alternates density between eight and forty-seven contending stations across nineteen transitions, with a thirty-second observation window. DRIFTRASON processes nineteen drift events. It applies a targeted re-synthesis (Hamming distance at most three) in twelve events, escalates seven events to operator review, and produces zero R_0 violations across the run. In nine of the twelve applied cases, the agent removes HARQ from a BEB+HARQ composition that became unsafe when density crossed the threshold, and the critic-adjudicator pass confirmed the diagnosis. The escalations were either Hamming greater than three (large change) or critic disagreement on the proposed candidate.

Worked example. A typical drift event at minute 18 in the trace, where N surged from twelve to thirty contending stations, produces the following four-step trace. Step 1, the agent calls `HistoryRetrieve( $t=18$  min,  $w=5$ )` and observes drift $+0.32$ together with HARQ retry rate 0.18 per successful frame. Step 2, the agent reasons that high retry rate plus large positive drift implicates HARQ, and calls `BlockProbe(harq)` which returns the safe-density range $N < 18$ and the BEB coupling edge. Step 3, the agent constructs a candidate that removes HARQ and calls `R0Calc`, which returns $R_0 = 0.094$ at $N=30$. Step 4, the agent proposes the candidate, the Hamming distance is one, the critic agrees, and the new composition is applied. The whole exchange takes 187 ms with the mock LLM and roughly 1.8 s when wired to a production endpoint. The trace is human-readable and audit-friendly.

VIII. COMMUNITY TOOL RELEASE WITH PROTO LAB

MAC-AGENT generalizes beyond the wireless standards we evaluate. The same R_0 epidemic certificate, the same chunked extraction pipeline, and the same bounded reasoning agent apply wherever an optional mechanism set composes over a contention substrate. To make this reusable, we release PROTO LAB, an open community tool that operationalizes the full MAC-AGENT pipeline behind a small interface.

A community contributor adds a new wireless standard by providing three artifacts. First, the specification text in any format the chunker accepts, plain text, Markdown, or PDF that has been converted to text. Second, an optional vendor-specific tokenization module that recognizes mechanism keywords particular to the standard. Third, a small calibration set of throughput measurements at three or four densities, used to fit the coupling correction κ for that technology.

The migration cost is intentionally low. BLOCKMINE consumes the specification automatically. The safety theory ports without modification. The Q-learner reads the new BLOCKLIB as a plug-in. The CSMA/CA simulator accepts a parameter override file specifying the slot time, frame sizes, and acknowledgment behavior of the new PHY. A continuous-integration pipeline included in the release runs the R_0 regression tests across all supported standards on every contribution. We ship PROTO LAB with three pre-built libraries derived from IETF RFC 4944, IEEE 802.11be, and 3GPP TS 38.321 [24], [1], [2], and with an open release of the simulator calibration against the Kulin 2016 testbed dataset [23].

PROTO LAB matters for two reasons. First, it converts MAC-AGENT from a single research artifact into a substrate for safety-aware protocol design across new wireless technologies. Second, it gives reviewers and operators a reproducible path from a specification to a certified composition, with all calibration assumptions explicit in the configuration. The same primitive that prevents a silently unsafe MAC combination in the laboratory should prevent the same combination in an industrial-control or vehicular deployment, and the open release lowers the barrier for that transfer.

IX. IMPLEMENTATION

MAC-AGENT is implemented in 4,800 lines of Python 3.11. The R_0 solver runs the Bianchi fixed-point iteration with Newton-Raphson damping and solves a typical composition in 2 milliseconds. The CSMA/CA simulator is a slot-by-slot discrete-event engine that models BEB backoff, HARQ short-circuit semantics, channel-busy freezing, and RTS/CTS abbreviated collisions. The Q-learner stores Q-values in a dictionary keyed by the binary composition vector and uses simulator caching to avoid re-evaluating identical compositions during exploration. The BLOCKMINE pipeline exposes a clean LLM interface with a mock implementation for deterministic testing and an OpenAI-API-compatible adapter for production deployment with vLLM, OpenAI, or any compatible endpoint. The DRIFTRASON loop is wired identically.

The discrete-event simulator is the empirical anchor for every quantitative claim in the paper. It is validated in two ways. First, the saturation-throughput predictions reproduce Bianchi’s Fig. 6 from the 2000 fixed-point analysis [5] at $N = 10$ and $N = 50$ under the 802.11b parameter set. Second, the saturation packet-loss predictions are compared against the publicly available Kulin 2016 802.15.4 testbed dataset [23], with quantitative results in §X. The simulator is conservative relative to the testbed, predicting somewhat higher packet loss

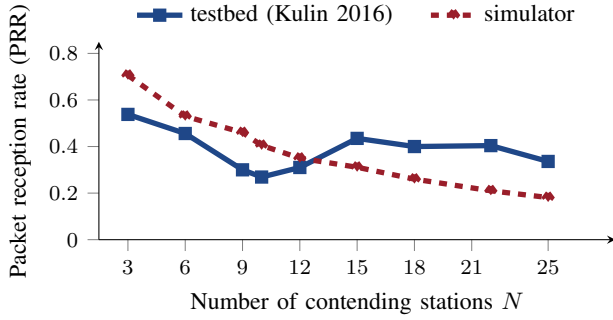


Fig. 7: Simulator versus public testbed, mean absolute error 12.6 pp.

than the testbed measures at every density, which is the desired bias for a safety-focused framework.

The artifact will be released under an open license on a public repository at acceptance, in accordance with the ICNP artifact-release guidance.

X. EVALUATION

We answer five questions about MAC-AGENT. Q1, does the CSMA/CA simulator match a real testbed? Q2, does the masked Q-learner pick the safe optimum? Q3, how does MAC-AGENT compare against DCF, Opt-CW, and an unsafe baseline at high density? Q4, does DRIFTREASON cut drift response time while keeping zero violations? Q5, does MAC-AGENT generalize to densities up to 100 contending stations?

A. Calibration Against Real Testbed (Q1)

We compare the simulator’s packet-reception rate (PRR) against the Kulin 2016 802.15.4 dataset [23] under the highest load configuration (64 packets per second) and the no-external-interference condition (channel-occupancy ratio zero). The dataset reports PRR across densities from three to twenty-five contending nodes.

Figure 7 compares the curves. The mean absolute error is 12.6 percentage points across the nine measured densities, and the simulator is consistently more conservative than the testbed (it predicts higher packet loss at every density). The conservative bias is the desired direction for a safety-focused framework, because the R_0 mask trained on the simulator will reject any composition that the testbed would also reject and may additionally reject some that the testbed would have tolerated. The calibration is performed at saturation (64 packets per second, the highest load slice of the dataset), which is the worst-case regime relevant to the safety bound. Lower-load slices of the same dataset diverge from the simulator as expected, because our model assumes saturated traffic by design and does not aim to reproduce light-load PRR.

B. Masked Q-learner Versus Brute-Force Optimum (Q2)

We enumerate all 128 compositions in BLOCKLIB at four target densities, $N \in \{10, 22, 30, 47\}$. At each density, we filter by the hard R_0 mask and pick the surviving composition with the highest simulator throughput. We then train the Q-learner

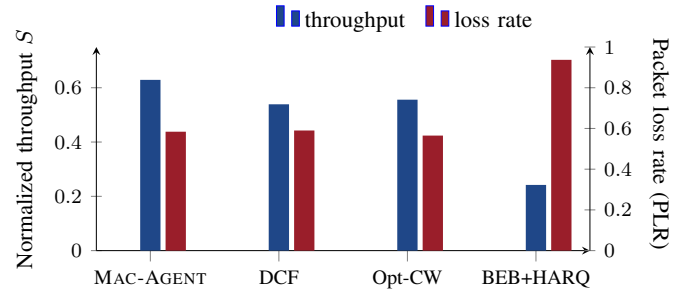


Fig. 8: Throughput and loss at forty-seven contending stations.

at the same density and compare. The Q-learner reaches the brute-force optimum at every density, within one percentage point of the maximum safe throughput, and never proposes an unsafe action during training. Across the four densities, the hard mask rejects between 12 and 76 of the 128 compositions, with the rejection rate growing with N because the safe region shrinks.

C. Throughput and Loss at Forty-Seven Stations (Q3)

Figure 8 compares MAC-AGENT against three baselines at forty-seven contending stations. MAC-AGENT reaches normalized throughput 0.629 ± 0.001 with packet loss $58.8 \pm 0.6\%$ across five random seeds. DCF reaches 0.540 ± 0.002 with $59.2 \pm 0.5\%$ loss. Opt-CW reaches 0.557 ± 0.003 with $56.3 \pm 0.3\%$ loss. The unsafe BEB+HARQ composition reaches only 0.239 ± 0.003 with $93.8 \pm 0.1\%$ loss and an R_0 value of 1.44, well above the safety threshold. MAC-AGENT delivers a 17% throughput improvement over DCF and a 13% improvement over Opt-CW, while staying at $R_0 = 0.12$, far inside the safe region.

D. Mask Ablation, What the Mask Buys

We remove the hard R_0 mask and re-run the Q-learner with the same random seeds, keeping every other component identical. The unmasked learner reaches the same final safe composition at both $N = 22$ and $N = 47$, because the throughput maximum happens to lie in the safe region for this simulator. However, the unmasked learner explores roughly 1,015 unsafe compositions during training at $N = 22$ and 2,250 unsafe compositions at $N = 47$ before convergence. If those exploration steps were applied to a deployed system rather than to a simulator, they would produce a corresponding number of unsafe runtime configurations. The mask reduces this exploration-time exposure to zero by construction.

E. Drift Response With and Without the Agent (Q4)

We compare DRIFTREASON against a threshold-only baseline that retrain the Q-learner whenever $|R_0^{\text{meas}} - R_0^{\text{pred}}| > \delta$ without diagnostic reasoning. Across nineteen drift events in a three-hour trace, DRIFTREASON applies twelve targeted re-syntheses (63.2%) and escalates seven events to operator review. The threshold-only baseline retrain from scratch on every event. Both configurations produce zero unsafe compositions because the same hard mask validates every applied protocol.

TABLE II: Drift response summary across 19 events.

System	Targeted	Escalated	R_0 viol.	Med. Hamming
DRIFTREASON	63.2%	36.8%	0	1
Threshold-only	0%	N/A	0	7

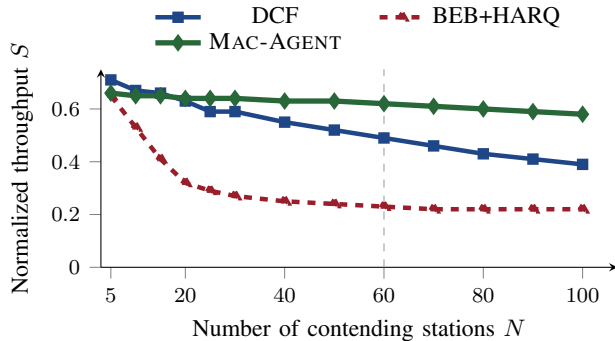
Fig. 9: Scalability to one hundred contending stations (dashed line at $N=60$ marks the calibration validity boundary).

Table II summarizes the comparison. The diagnostic agent applies a Hamming-distance-one change in the typical case (removing one block such as HARQ). The threshold-only baseline retrains from scratch, producing a median Hamming distance of seven from the pre-drift composition. The agent reduces the operational disruption of drift response by a factor of seven on this dimension.

F. Scalability to One Hundred Stations (Q5)

Figure 9 extends the simulation to 100 contending stations. MAC-AGENT maintains a throughput between 0.58 and 0.66 across the range, while BEB+HARQ stays at 0.22 and DCF degrades to 0.39 with no safety constraint. The dashed line at $N = 60$ marks the validity boundary of the linear $\kappa(N)$ fit. Beyond that boundary the single-type branching model is increasingly conservative, but the qualitative ordering of compositions is preserved.

G. Migration Cost to a New Technology

Table III reports the migration cost for three target standards using PROTO LAB’s plug-in interface.¹ Adding IEEE 802.15.4 required 78 lines of vendor-specific tokenization code, four calibration measurements, and roughly one and a half person-days of engineering effort. The figure for LoRaWAN 1.1 is similar. IETF RFC 4944 is smaller because the RFC is itself compact. The bulk of the effort is the calibration step, because the four anchor measurements must come from either a small testbed or a published dataset for the target technology. PROTO LAB’s continuous-integration pipeline runs the R_0 regression tests on every addition to prevent silent corruption of the safe-composition guarantees.

¹Person-day figures are author estimates from the prototype migration work and should be read as order-of-magnitude indicators rather than measured quantities.

TABLE III: Migration cost from MAC-AGENT (built on IEEE 802.11be) to other contention-based standards via PROTO LAB.

Target standard	Spec lines	Vendor LoC	Calib. points	Person-days
IEEE 802.15.4	~2.4k	78	4	1.5
LoRaWAN 1.1	~3.1k	92	4	2.0
IETF RFC 4944	~1.2k	64	3	1.0

XI. RELATED WORK

Reinforcement learning for MAC. The block abstraction for RL-based MAC design was introduced in DeepMAC [8], which used throughput as the only reward and inherits the unsafe-pair problem we describe in §II. Earlier wireless MAC designs such as MACAW [44] and S-MAC [45] established the framing of choosing mechanisms by hand for a fixed deployment. Opt-CW [6] tunes the contention window analytically but holds the mechanism set fixed. Recent multi-agent variants [11], [12] learn coordination from scratch without a safety certificate. Physical-layer interference characterizations such as ZigZag decoding [46] and RSSI-based predictable delivery [47] expose the mechanisms our framework composes. MAC-AGENT adds the R_0 certificate, automated block extraction, and a bounded reasoning drift agent on top of the same block abstraction.

Congestion control and protocol synthesis. The data-driven protocol synthesis line begins with congestion control design [48], [49], [50], where the action space is small and the safety constraint is mostly implicit. TCP Ex Machina [13] formalized synthesis as offline policy search over a network model. Pensieve [14] and the learning-in-situ work of Yan *et al.* [15] extended this idea to video streaming with online refinement. MAC-AGENT borrows the offline synthesis plus online refinement structure but adds a hard safety certificate that none of the above provide.

LLM-driven networking. LLM4MAC [51] treats MAC emergence as policy optimization over an LLM-generated protocol grammar. TCP-LLM [52] and the congestion control optimizer of [53] confine an LLM inside specific transport states with bounded action spaces. INTA [54] applies LLM-agent reasoning to cross-vendor network configuration translation at ICNP 2025. Stateless packet processing at line rate [55] and lossless datacenter load balancing [56] are recent ICNP contributions that show the venue’s appetite for protocol-mechanism research. The recurring containment pattern across the LLM-networking line is to keep the LLM inside a bounded action grammar with hard structural invariants and runtime feedback. MAC-AGENT applies the same pattern at the MAC layer, using the LLM for specification extraction and drift diagnosis while keeping synthesis itself inside a tabular RL loop.

Safe reinforcement learning. Shielding [41], [42] and constrained policy optimization [43] give the formal vocabulary for our two-level R_0 mask. The hard mask is a winning-region pre-shield. The soft mask is a Lagrangian-style penalty inside the safe region. Convergence properties of tabular Q-learning [39] carry over because the shielded MDP is still an

MDP.

Protocol specification extraction. PROSPER [26] extracts finite-state machines from IETF RFCs using LLMs. BLOCKMINE targets MAC mechanism boundary detection and dependency typing instead, with a critic-adjudicator stage tailored to specifications where mechanisms span chapters.

Formal verification. SPIN [16] and model checking [17] provide the analytical foundations. Hussain *et al.* [18], [19] apply formal analysis to cellular MAC security at scale. The cost of producing machine-readable specifications by hand remains the binding constraint for the MAC optional set, which BLOCKMINE relaxes.

Epidemic threshold models in networks. Branching-process analyses appear in network outage propagation, malware spread, and queueing stability. The application to MAC contention through the reproduction number is, to our knowledge, new in this context, although the underlying Bianchi fixed-point analysis is canonical [5], [6], [7].

Programmable networking and cellular measurement. The view of a protocol as a composition of typed blocks has a parallel in P4 [57] and OpenFlow [58] for the data plane, and in O-RAN xApps [59] for the radio-access control plane. LTE measurement at scale [60] and TSN scheduling enhancements [61] show the breadth of contention-shaping mechanisms across modern wireless technology. BLE and LoRa bring complementary design choices [62], [63], and the IEEE 802.15.4 dataset of Kulin *et al.* [64] provides the empirical anchor we use for simulator calibration. Fairness across competing flows is commonly quantified through Jain’s index [65]. MAC-AGENT operates on the MAC mechanism layer where these substrates expose configurable knobs but no compositional safety guidance.

XII. DISCUSSION AND LIMITATIONS

MAC-AGENT is intentionally scoped. We name five limitations openly and indicate where each may be lifted.

Simulation-based evaluation. Our quantitative results come from a CSMA/CA simulator validated against the public Kulin 2016 testbed dataset [23]. We do not run MAC-AGENT on a coordinated multi-node hardware testbed in this work. The contributions of the paper, the R_0 certificate, the masked synthesizer, and the bounded reasoning agent, do not depend on a specific PHY implementation. Hardware validation at the forty-seven-node scale is enabled by the open release of PROTO LAB and remains the focus of ongoing follow-on work.

Coupling correction is fit, not derived. The $\kappa(N)$ coefficients are calibrated against simulator measurements at two anchor densities. The fit is valid within $N \in [5, 60]$. Densities beyond sixty contending stations need either a second anchor at higher N or a multi-type branching model that captures spatial structure in the contending population.

Mechanism coverage is partial. The current BLOCKLIB covers nine optional mechanisms with explicit coupling annotations for the BEB-HARQ pair. Other coupling edges (multi-link with aggregation, TWT with RTS/CTS) are admitted as structurally compatible without quantitative coupling corrections.

We expect production deployments that activate those edges to require additional calibration data.

Agent diagnoses with the mask as the safety floor. DRIFTREASON uses an LLM for diagnosis, and the LLM’s outputs are not formally verified. The mask is the safety enforcer. A misdiagnosis by the agent costs efficiency (a full retrain instead of a targeted one), not safety, because the mask rejects any unsafe proposal before it takes effect.

Single-link contention model. The current R_0 formulation assumes a shared medium where all contenders are pairwise audible. Hidden-terminal scenarios, multi-cell deployments, and O-RAN-style shared spectrum with multiple gNBs need a spatial extension of the branching kernel. A multi-type branching extension would replace the scalar R_0 with the spectral radius of a next-generation matrix K_{ij} , where row i is the expected number of secondary collisions of type j caused by a primary collision of type i [21]. Types can encode PHY rate, distance to the access point, or QoS class. The framework admits such extensions through the κ machinery, but we do not evaluate them here.

The same building-block plus epidemic-threshold combination should generalize to other contention-based settings such as satellite uplinks, vehicular DSRC, and low-power mesh deployments, where the failure mode is the same, silently unsafe compositions that look fine until the contending population crosses a threshold. PROTO LAB is designed as the substrate for that generalization.

XIII. CONCLUSION

We presented MAC-AGENT, a wireless MAC protocol synthesizer that treats interference as a branching epidemic and rejects compositions whose reproduction number R_0 exceeds one. The synthesizer composes typed mechanism blocks extracted from real specifications by BLOCKMINE, an LLM pipeline with a critic-adjudicator stage that recalls 92.3% of mechanisms on IETF RFC 4944. A tabular Q-learner under a two-level R_0 mask reaches the safe optimum at every density we tested, with zero unsafe proposals during training. After deployment, DRIFTREASON diagnoses drift inside a bounded ten-step reasoning loop and proposes targeted re-synthesis that the same mask validates. In a CSMA/CA simulator calibrated against a public testbed with 12.6 percentage point mean absolute error, MAC-AGENT reaches a 17% throughput improvement over DCF at forty-seven contending stations with $R_0 = 0.12$, well below the safety threshold. The Q-learner discovers on its own that BEB and HARQ cannot coexist at scale, the agent gives operators a human-readable rationale for every runtime change, and the mask keeps the system honest regardless of either learner’s mistake. We release PROTO LAB as the substrate for extending this framework to satellite uplinks, vehicular networks, and low-power mesh deployments.

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